

Revisiting a Pain in the Neck: A Semantic Reasoning Benchmark for Language Models

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Motivation & Research Question

“A blind spot in LM evaluation”

LMs are benchmarked on math, code, and logic — but rarely on **fine-grained semantic reasoning** over phrases. **Semantic phrases** (multiword expressions) carry meaning that is *not* derivable from their parts, yet existing evaluations use **isolated task formats**, so strong scores may reflect task-specific heuristics rather than stable understanding.

How do language models behave on phrasal semantics across distinct but structurally constrained task operations?

We take an **operation-aligned** view: instead of scoring one task, we test whether phrasal understanding *transfers* across operations that share the same sense but differ in output constraints.

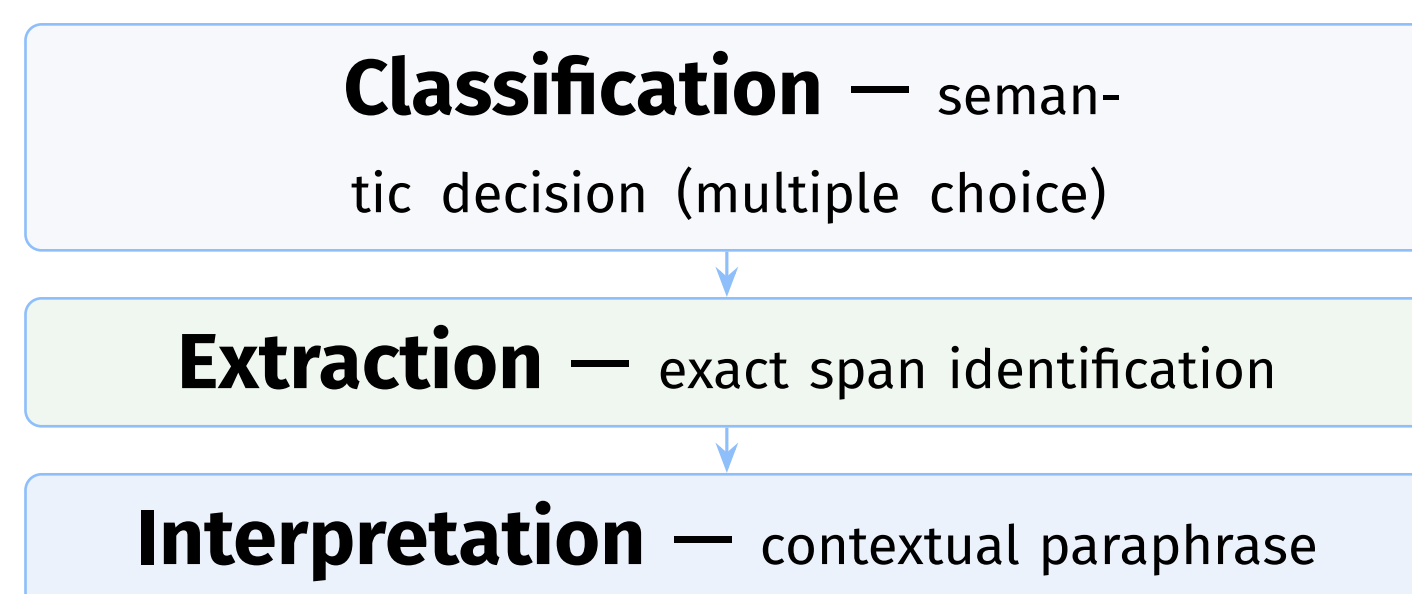
Phrase Types & Atomic Operations

“Four phenomena, three probes”

Four phrase types span the compositionality spectrum:

- **LC** — Lexical Collocations (*broadly compositional*)
- **IE** — Idiomatic Expressions (*e.g. “kick the bucket”*)
- **NC** — Noun Compounds (*“baby/olive oil”*)
- **VC** — Verbal Constructions (*LVC / VPC / VID*)

Three atomic operations probe the same sense under different constraints:



Formulation. SP processing is cast as constrained generation: given prompt $\mathcal{P}$ and contextualized phrase $\mathcal{S}$, the model produces $\mathcal{O}$ from $\mathcal{I} := \mathcal{P} \oplus \mathcal{S}$, with $\mathcal{O}$ shaped by the target operation.

Contributions

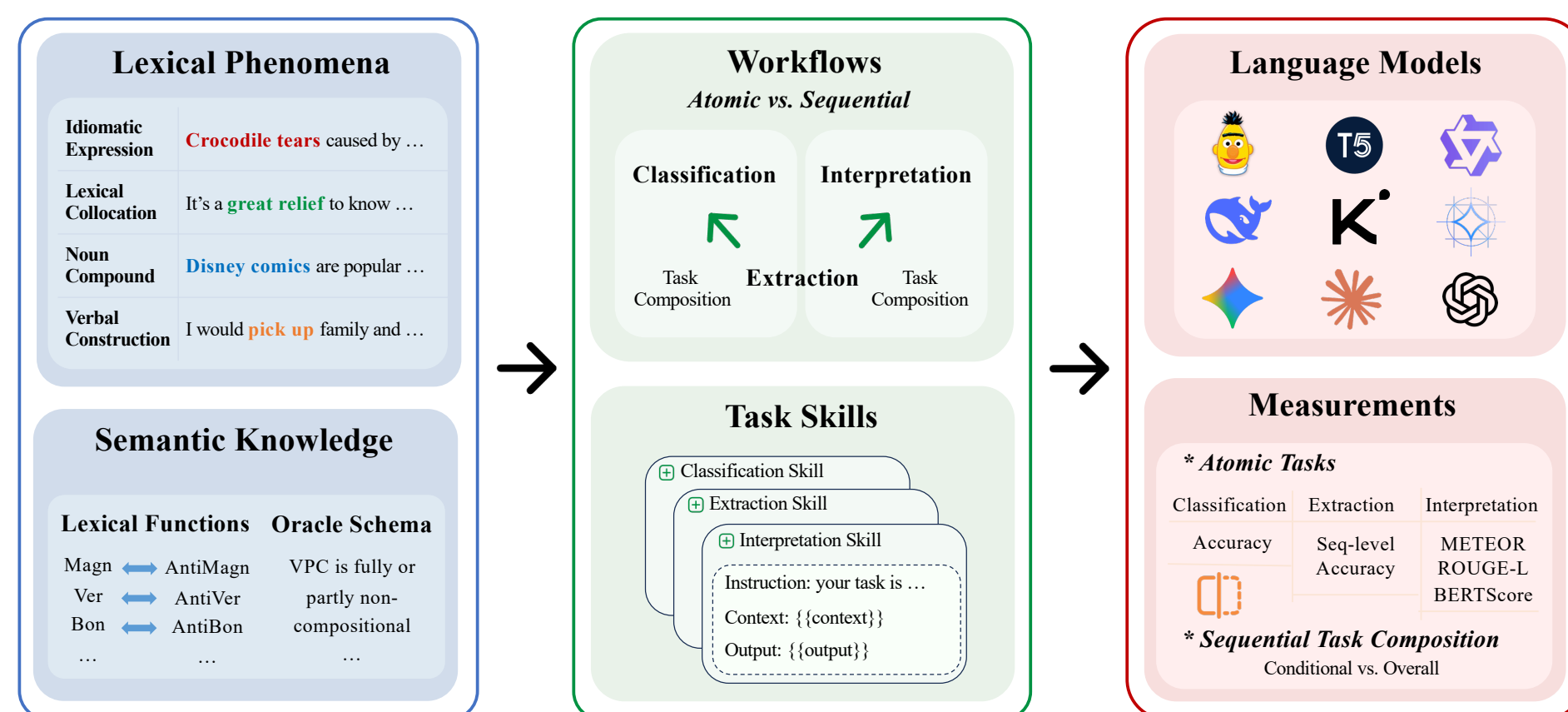
“What SEMANTICQA adds”

- **Operation-aligned evaluation.** We align SP tasks with the semantic operations they instantiate, enabling cross-family analysis.
- **Minimal, controlled design.** Fixed prompt templates isolate the operation as the source of performance gaps.
- **Cascade diagnostics.** In sequential setups, strong LMs **lose semantic consistency** — hidden by single-task evaluation.
- **Broad model coverage.** 10+ open-source and proprietary LMs benchmarked against a human reference.

The SEMANTICQA Evals Framework

“A unified, operation-aligned testbed”

SEMANTICQA consolidates prior MWE resources into a **unified, operation-aligned testbed**. It is *diagnostic*, not a leaderboard: the focus is within-task trends and relative changes induced by operations and compositions.



Phrase types $\times$ operations $\times$ diverse LMs, evaluated alone and in sequential compositions.

Exemplar (Idiom “rocket science”). The same sense is probed three ways: **Interpretation** $\Rightarrow$ “something complex and difficult”; **Extraction** $\Rightarrow$ span “rocket science”; **Classification** $\Rightarrow$ (B) **Difficult problem** ✓. Chaining them defines a **task composition**.

Main Benchmarking Results

“No model wins everywhere”

Performance varies **substantially** across operations and phrase types — **no model wins everywhere**. The benchmark is neither saturated nor uniformly hard.

	Idiom			Collocation			Noun Cmpd.			VMWE		
Model	IED	IEE	IEI	LCC	LCE	LCI	NCC	NCE	NCI	VPE	LVE	VIE
Human	71.0	87.0	20.5	47.0	50.0	16.7	71.0	73.0	17.2	85.0	55.0	78.0
DeepSeek-R1	84.3	72.3	19.2	76.4	64.3	33.6	62.7	70.7	68.7	81.6	35.8	59.1
Kimi-K2-Inst.	81.7	69.6	23.5	79.7	69.2	39.1	64.7	64.4	76.7	81.1	43.3	46.7
Gemma-3-27B	72.1	62.0	19.9	70.8	68.2	38.7	58.3	59.2	70.5	74.1	35.0	52.4
Claude-Sonnet-4.5	78.0	72.0	26.7	77.1	72.7	41.2	70.1	62.1	83.8	82.0	37.2	47.6
OpenAI o3	83.5	77.4	21.9	85.9	71.5	41.6	65.2	78.6	74.5	77.3	36.9	52.4
GPT-5	85.4	78.7	22.6	86.2	71.5	37.4	67.2	79.0	75.3	74.7	38.3	56.2
Human ref.	Open-source	Proprietary	best	human > all LMs. Best of zero/few-shot shown; metrics: Acc / Acc _s / METEOR (%).								

Strengths are operation-specific, not a single transferable notion of phrase-level competence — behavior differs by operation even within one phenomenon.

Code, Data & Resources

“Come to get the released artifacts of SemanticQA!”

Scan for Website

Code & Data:
github.com/jacklanda/SemanticQA

Dataset:
huggingface.co/datasets/jacklanda/SemanticQA

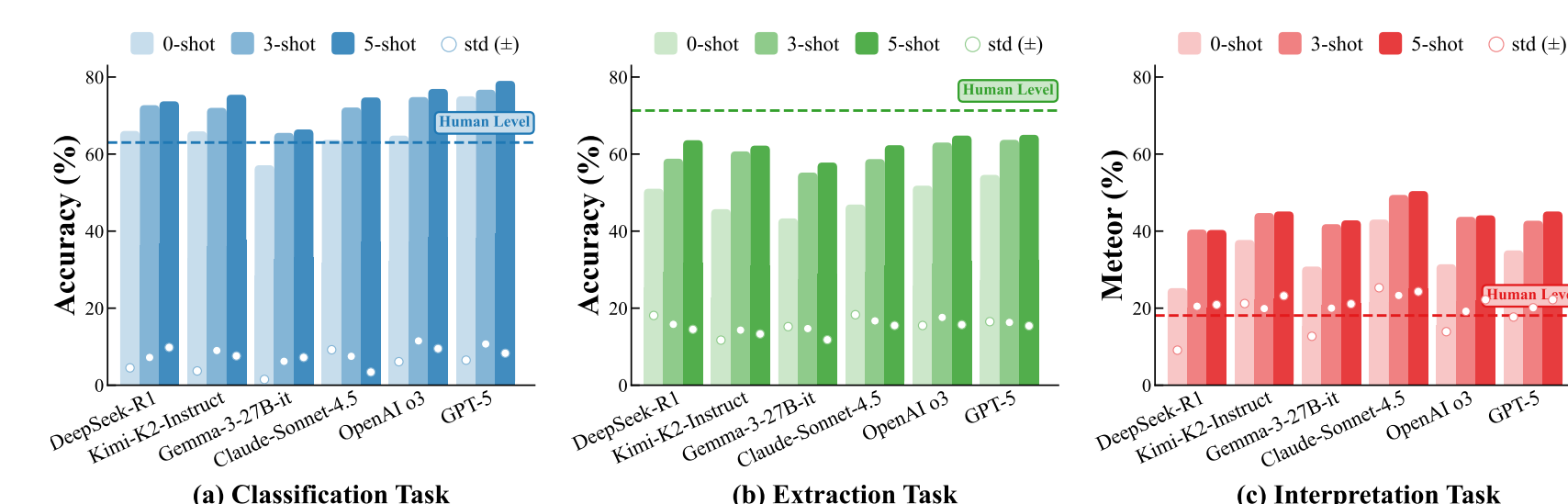
Slides:
github.com/SemanticQA/SemanticQA.github.io

Unified MWE testbed: 4 phrase types, 3 operations, 10 tasks.

In-Context Learning & Category Scaling

“Few-shot helps unevenly across operations”

- **Interpretation:** benefits **most consistently** from few-shot demos — but largely exemplar-guided reconstruction, not semantic grounding.
- **Classification:** mixed — weak zero-shot models improve; others plateau or **regress**.
- **Extraction:** **least stable** — gains when span structure is shown, degrades when test instances diverge.



Bars: mean per-category performance; markers: std across tasks.

Scaling. On LCC, accuracy **drops as categories grow** (1 $\rightarrow$ 16). Supervised baselines stay stable; LLMs drop **sharply** (DeepSeek-R1: 81.7% $\rightarrow$ 35.4%) — in-context reasoning does *not* replace explicit supervision at high relational granularity.

Sequential Task Compositions

“Extraction is the bottleneck”

Dependent operations in sequence (*extract* $\rightarrow$ *interpret*). Conditional METEOR (on correctly extracted phrases) far exceeds overall scores $\Rightarrow$ **extraction is the bottleneck**: fluent interpretation **cannot** compensate for upstream errors.

Type	Model	Ext.	Cond.	Overall
LC	DeepSeek-R1	33.8	42.3	14.3
	GPT-5	41.3	41.8	17.3
IE	DeepSeek-R1	51.3	12.0	6.2

Compositional classification degrades sharply at 8- and 16-class settings; few-shot only *partially* reduces the dependence on extraction quality.

Key Takeaways

“What to remember”

- **Multi-dimensional evaluation is needed.** Strengths are **operation-specific**; no single task or metric captures phrasal competence.
- **Metrics shape apparent strengths.** Similarity metrics reward paraphrasing; strict span metrics expose **structural brittleness**.
- **Workflow robustness is limited.** Errors **propagate** through compositions; extraction is the dominant bottleneck.
- **Scaling does not rescue reasoning.** Accuracy **collapses** as relational granularity grows — no substitute for supervision.

SEMANTICQA is a diagnostic lens, not a leaderboard: it reveals where and why phrasal understanding breaks down.

Acknowledgement

- In loving memory of the late Prof. **Igor Mel'čuk**, whose seminal work on *lexical functions*, *Meaning-Text Theory*, and the broader field of lexical semantics is the most direct inspiration for SEMANTICQA. Though we never met him in person, we have long felt a kinship of the mind and hold his vision in deep admiration.
- We thank the anonymous reviewers for their constructive feedback, and the maintainers of the underlying MWE resources, whose datasets made the unified SEMANTICQA testbed possible.
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